

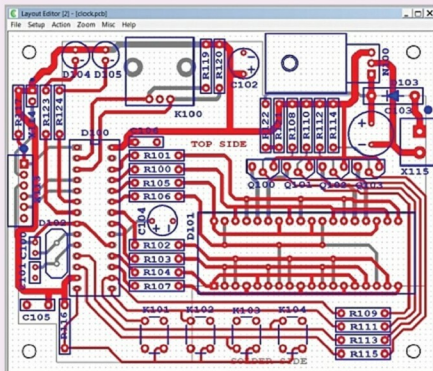


Fig. 2: A PCB layout drawn with CometCAD

This is called a cut-out and can be milled in the final stage.

Architecting the board

You can choose to create a new PCB layout from scratch or update an existing PCB file, which is called forward annotation. Every PCB layout requires a circuit schematic as an input, and on importing the same, CometCAD allows you to estimate the area of the final PCB.

The layout editor allows you to edit the different layers of the PCB. The tool provides a polygon function, which can be used to set the borderline for your board and include gaps or cuts and copper planes on the PCB. These parameters will decide the fill rate area for the PCB. In other words, these will decide the area of the PCB that is available for the parts to be placed.

For making a selection, say a footprint, the layer you choose is the active layer. Setting the active layer is in your hands. You can also change the width of a trace along the PCB, although it cannot be

smaller than that specified in the rules. You can add parallel traces to change a specific route and also divide a trace into new segments.

Vias, which provide contact between metals of different layers, cannot be moved once connected. While a trace can run from the pad of a footprint to a via, an unconnected via cannot have a trace connecting it to a pad.

Find with ease. It is always difficult to identify specific components once the design is almost complete, thanks to the mess of traces and layers. A well-managed find option here lets you zoom to the exact point you are looking for, be it a node, a signal or a certain resistor. This shows up along with the traces connected to it, making it easy for you to debug.

Putting together a circuit

It is, of course, the circuit schematic that decides how your setup functions. The schematic is created on a user-specified sheet frame, and each action performed is considered as a

function by the tool. General basic options like create, move, mirror and rotate can be accessed by the given menus or shortcut keys. Each component can be configured according to your needs and can be appropriately named.

Complex designs like a communication interface demand a bus, which is a simple graphics item on the tool, but connections will have to be made manually. While moving a symbol does not automatically move the wires connected to it, a function called fix wires can be exploited to help with this aspect. The function can not only remove unwanted wires but can also split wires into segments that can be selected separately. The tool can even generate rectangular wire corners wherever specified.

Creating the basic element, a symbol

A symbol editor takes care of your requirements for creating the symbols that make up your design. A symbol could be an ordinary, graphics or connector type. While regular connection of wires is the norm, graphics symbols take care of graphics and even text that you may want to include in your circuit. With an automatic check of every pin number you enter for the symbols, any number that is missed out is flagged to the user immediately.

Configurable connectors aid achieving connection between dif-

Requirements for a PCB

It is important to make sure that every symbol has a different part name, so that the tool does not get confused during instantiation. While symbols with different number of pins can refer to the same footprint, like power pins, pin strings that refer to these have to be different, in accordance with the rules of the package section. These errors show up when a design is transferred to a PCB.